

YASH RAJ PANDEY

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Summary

AI engineer specializing in local-first LLM systems: agent runtimes, RAG and tool-calling platforms, and the production evaluation frameworks that keep them honest. I benchmark frontier and open-weight models, deploy self-hosted inference on-premise, and gate every release behind regression evals. Promoted from Software Engineer to AI Agents Architect in 13 months; 45+ merged PRs across 30+ open-source projects (llama.cpp, Apple MLX, Google Research).

Experience

University of Florida

Gainesville, FL

AI Agents Architect

Apr 2026 – Present

- Re-platforming inference onto institutional GPU infrastructure, self-hosting a ~745B-parameter open-weight MoE (FP8, 1M-token context) to cut projected 5-year TCO ~60% versus commercial APIs and per-query cost to ~\$0.01, fully on-premise.
- Architecting and deploying a production AI agent platform pairing RAG over a 10,000+ document technical corpus with an agentic tool-calling runtime that executes multi-step workflows autonomously, translating researchers' scientific workflows into natural-language access to domain data and analysis.
- Building a model-selection framework benchmarking 14 LLMs across 26 task-specific evaluations and 5 capability dimensions, identifying the top performer (88% multi-tool orchestration) and flagging hallucination failure modes before deployment.
- Building and maintaining a 24-case regression eval suite that gates every model and pipeline change for zero-regression backend swaps, and mentoring engineers on the evaluation methodology behind it.
- Hardening the data foundation behind the platform: migrating a 50,000+ record corpus into a typed analytical store (DuckDB), auditing 1.4M+ cells with zero mismatches and surfacing 2,000+ hidden ID conflicts before lock-in.

University of Florida

Gainesville, FL

Lead Software Engineer

Oct 2025 – Apr 2026

- Led and scaled a genomics data platform (Django, React, PostgreSQL) to 5.38M+ records across 32 data models, 58 endpoints, and 35 indexes; it became the system of record for 30+ researchers across 5 labs (+40% daily active users).
- Tuned PostgreSQL indexing and caching to hold trait lookups at 23–29 ms and analytics aggregations at ~222 ms over ~10K records, with BLASTN genome searches at ~602 ms on a 518 MB index.
- Shipped 7 ingestion and validation pipelines with automated R Markdown reporting (preprocessing from hours to ~2 minutes); deployed on GCP (Kubernetes, Terraform, Docker) with JWT/RBAC, cut frontend load time from 8s to 3s, and grew the team from one to three.

Software Engineer

Mar 2025 – Oct 2025

- Prototyped the original PostgreSQL schemas and Python ingestion; wrote the validation checks the production system kept.

Open-Source Projects

Looma / *Local-first memory for coding agents*

github.com/devYRPauli/looma

- Designed and built a CLI that converts Claude Code, Codex, and Cursor transcripts into structured work items and token-budgeted context packs, on the Python standard library alone (SQLite + FTS5, zero dependencies); 134 passing tests.
- Designed and benchmarked the extraction pipeline to F1 0.86 (0.94 precision) with a dependency-free heuristic and an optional local-LLM extractor reaching 0.95; optimized the ingest daemon ~27x and improved retrieval ranking (MRR +49%).

mddocs / *Git-native collaborative Markdown with an agent API*

github.com/devYRPauli/mddocs

- Built and published (npm, MIT) a local-first, git-native collaborative Markdown editor: real-time multiplayer (Yjs CRDT) plus a token-gated agent HTTP API with an SSE event stream and rate-limit headers; shipped 8 semver releases and cut package size 87% with esbuild.

Independent Evaluations / *TurboQuant KV-cache compression and Google TabFM*

yashrajpandey.com/writing

- Took TurboQuant (Google Research, ICLR 2026) from 0% to 100% needle retrieval at 16K tokens on a 16 GB M1 Pro with 3.5x less KV-cache memory; five root-caused fixes across MLX and llama.cpp forks, contributed upstream.
- Independently reproduced Google's TabFM across 3 machines and 13 TabArena datasets, beating Optuna-tuned XGBoost on all 10 fold-matched datasets; root-caused a multi-GPU predict crash, fix merged into google-research/tabfm.

Technical Stack

AI & ML:

vLLM, Ollama, llama.cpp, MLX, RAG, Vector Search, Agent Runtimes, Tool-Calling, Model Benchmarking & Evals, Self-Hosted MoE Inference (FP8), Quantization, KV-Cache, Reranking, Embeddings, MCP, LangChain, LangGraph, LlamaIndex, OpenAI & Anthropic APIs

Languages:

Python, JavaScript, TypeScript, SQL, Bash, C/C++

Frameworks:

Django, FastAPI, React, Next.js, Node.js

Databases:

PostgreSQL, DuckDB, Qdrant, ChromaDB, Redis, SQLite

Infrastructure:

Docker, Kubernetes, Terraform, GCP, AWS, Linux, GPU / Self-Hosted Inference, Apple Silicon / MLX, Git, GitHub Actions, JWT / RBAC

Education

University of Florida

Aug 2023 – Dec 2024

M.S. Computer & Information Science & Engineering (GPA 3.8)

Gainesville, FL

Jaypee University of Engineering and Technology

Jul 2019 – May 2023

B.Tech. in Computer Science and Engineering

Guna, India